The RHF and CASSCF analytic nuclear Hessians

design notes: the skeleton Fock derivatives, CP-CASSCF and the state average

free-breakfast design notes

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Abstract

Part I is the closed-shell RHF Hessian's notation and its skeleton Fock derivatives, the $3N$ AO matrices $F^{(x)}$ every coupled-perturbed right-hand side needs, and with them the one-electron derivative integrals as matrices rather than contracted scalars. Part II is the CASSCF Hessian, state-specific and state-averaged: the second-order connection, the CP-CASSCF right-hand side and its $3N$ solves with the Z-vector's operator, the state average's Lagrangian, whose second derivative needs the gradient's own Z and a third derivative of the averaged energy contracted once with it, and the rotations among the averaged roots. Both are excerpts: the parts of the design that the module READMEs cite, with the notation that makes them read.

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Part I
RHF

1 Notation and the energy

Orbitals are real and the reference is closed-shell RHF with n_o doubly occupied orbitals. AO indices are $\mu\nu\lambda\sigma$, occupied MOs $ijkl$, virtual MOs $abcd$, and general MOs $pqrs$. C is the MO coefficient matrix, C_o and C_v its occupied and virtual columns, and ε_p the canonical orbital energies. Nuclear coordinates are x and y; each is one Cartesian component of one nucleus, $x = (A, k)$, and there are $3N$ of them for N atoms.

The density and energy-weighted density are

$$D = 2\,C_o C_o^{\mathsf{T}}, \qquad W = 2\,C_o\,\varepsilon_o\,C_o^{\mathsf{T}} = \tfrac{1}{2}\,DFD, \tag{1}$$

the second form of W holding for any S-orthonormal C that diagonalizes the occupied block of F (since $C_o^{\mathsf{T}} F C_o = \varepsilon_o$). It is the form that differentiates cleanly.

With the antisymmetrized integral

$$(\mu\nu\|\lambda\sigma) = (\mu\nu|\lambda\sigma) - \tfrac{1}{2}(\mu\lambda|\nu\sigma), \tag{2}$$

which is symmetric under $(\mu\nu) \leftrightarrow (\lambda\sigma)$ for real functions, the two-electron operator on a symmetric density P is

$$G(P)_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma}(\mu\nu\|\lambda\sigma) = J(P)_{\mu\nu} - \tfrac{1}{2}K(P)_{\mu\nu}, \tag{3}$$

and G is a symmetric bilinear form: $\mathrm{tr}[P\,G(Q)] = \mathrm{tr}[Q\,G(P)]$. The Fock matrix is $F = h + G(D)$ and the energy is

$$E = \sum_{\mu\nu} D_{\mu\nu} h_{\mu\nu} + \tfrac{1}{2} \sum_{\mu\nu\lambda\sigma} D_{\mu\nu} D_{\lambda\sigma} (\mu\nu\|\lambda\sigma) + V_{nn}. \tag{4}$$

The SCF conditions are $F_{ai} = 0$ and $C^{\mathsf{T}} S C = 1$, at every geometry.

Derivative integrals. A superscript on an integral means its derivative at fixed AO coefficients, the basis functions moving with their nuclei: $h^x_{\mu\nu} = \partial h_{\mu\nu}/\partial x$, S^x, and $(\mu\nu|\lambda\sigma)^x$; likewise h^{xy}, S^{xy} and $(\mu\nu|\lambda\sigma)^{xy}$. h^x includes the derivative of the nuclear-attraction *operator* as well as of the basis functions. MO transforms with the unperturbed coefficients are written $S^x_{pq} = C_p^{\mathsf{T}} S^x C_q$, and similarly for any AO matrix.

Skeleton Fock derivative. The derivative of F at fixed D,

$$F^{(x)}_{\mu\nu} = h^x_{\mu\nu} + \sum_{\lambda\sigma} D_{\lambda\sigma} (\mu\nu\|\lambda\sigma)^x, \tag{5}$$

is the central intermediate. There are $3N$ of them, each an AO matrix. The Hessian is an explicit part, the second-derivative integrals contracted with the converged density, plus a response part that needs the first-order orbital response, one coupled-perturbed Hartree–Fock (CPHF) solve per nuclear coordinate; $F^{(x)}$ enters both, and every CPHF right-hand side.

2 The skeleton Fock derivatives

Both forms of the Hessian, and every CPHF right-hand side, need the $3N$ AO matrices $F^{(x)}$ (5). The two-electron part is a *derivative J/K build*: $J^{(x)}(D)_{\mu\nu} = \sum D_{\lambda\sigma}(\mu\nu|\lambda\sigma)^x$ and the matching $K^{(x)}$, for every x at once. This is the kernel the tree most clearly lacks. Its first-derivative integrals exist (`coulomb_gradient` and `exchange_gradient` generate $(\mu\nu|\lambda\sigma)^x$), but they contract both index pairs with D and return $3N$ scalars. A Hessian needs the same integrals contracted with one D, returning $3N$ matrices: $3N\,n_{\mathrm{AO}}^2$ doubles of output, and scatter writes into per-atom slabs.

The one-electron part. h^x as matrices rather than scalars is the same change to `kinetic_gradient` and `potential_gradient`, and S^x as matrices to `overlap_gradient`; all three are cheap.

A shell quartet differentiated on the centre of shell μ contributes to $J^{(x)}$ through every $(\mu\nu)$ and $(\lambda\sigma)$ element it touches, so the natural layout is one n_{AO}^2 slab per coordinate, written atomically or reduced per block. With translational invariance, three of four centres suffice and the fourth's contribution is minus the sum. Memory: at 500 AOs and 30 atoms the $F^{(x)}$ alone are 180 MB, and the CPHF's responses and right-hand sides are small beside them.

Since written. The one-electron part is `overlap_derivative`, `kinetic_derivative` and `potential_derivative`, the $[n_{\mathrm{atom}}, 3, n_{\mathrm{AO}}, n_{\mathrm{AO}}]$ tensors of `src/integrals/one_electron/gradient`; the two-electron part, over density fitting, is `df_jk_derivative` (`src/integrals/eri/density_fitting/gradient`). `src/hessian` assembles $F^{(x)}$ from them.

Part II
CASSCF

3 Scope, notation and the energy

Orbitals are real, the active space is complete, and the CI vector lives in a fixed determinant basis. MO indices: core i, j; active t, u, v, w; virtual a, b; general p, q, r, s. AO indices $\mu\nu\lambda\sigma$; aux indices P, Q. Nuclear coordinates are x and y, $3N$ of them.

With the 1- and 2-RDMs D_{pq} and Γ_{pqrs} over all MOs (core blocks closed-shell, active blocks from the CI vector), the energy is

$$E = \sum_{pq} h_{pq} D_{pq} + \tfrac{1}{2} \sum_{pqrs} (pq|rs)\,\Gamma_{pqrs} + V_{nn}, \tag{6}$$

and the generalized Fock matrix, `response::fock`'s, is

$$F_{pq} = \sum_{r} h_{pr} D_{rq} + \sum_{rst} (pr|st)\,\Gamma_{qrst}. \tag{7}$$

Only its first n_{occ} columns are nonzero. The parameters are $\lambda = (\kappa, c)$: the non-redundant rotations κ (core–active, core–virtual, active–virtual), $C(\kappa) = C\,e^{K}$ with K antisymmetric, and the CI vector. Their gradient is the orbital gradient $g_{pq} = 2(F_{pq} - F_{qp})$ and the CI gradient $2P(H-E)c$, $P = 1 - cc^{\mathsf{T}}$. Both vanish at convergence.

Two energies are in scope:

- **state-specific** (SS): one root, E stationary in all of λ (section 6);
- **state-averaged** (SA): root I's energy E_I at the orbitals that make $E_{\rm SA} = \sum_r w_r E_r$ stationary, with every root an eigenvector of the same active-space Hamiltonian H_0 (section 7).

Derivative integrals. As in Part I (section 1), h^x, $(\mu\nu|\lambda\sigma)^x$, S^x and their second derivatives are taken at fixed AO coefficients, and transformed to the MO basis with the unperturbed C: $S^x_{pq} = C_p^{\mathsf T} S^x C_q$. A quantity in brackets, $F^{[x]}$, $H^{[x]}$, $g^{[x]}$, means the same expression as F, H, g evaluated at the derivative MO integrals $(h^x_{pq}, (pq|rs)^x)$ in place of $(h_{pq}, (pq|rs))$, at the unperturbed RDMs.

4 The gradient, as the tree computes it

`mcscf::df_gradient` (`src/mcscf/gradient/README.md`) computes

$$E^x = \sum D^{\rm AO}_{\mu\nu} h^x_{\mu\nu} + \tfrac{1}{2} \sum \Gamma^{\rm AO}_{\mu\nu\lambda\sigma} (\mu\nu|\lambda\sigma)^x - \sum W_{\mu\nu} S^x_{\mu\nu} + V^x_{nn}, \qquad W = C\ \mathrm{sym}(F)\, C^{\mathsf T}, \tag{8}$$

with the connection $C(x) = C(1 - \frac{1}{2} S^x_{\rm MO}\, \delta x)$, so that $\partial E/\partial U_{pq} = 2F_{pq}$ turns the connection into $-\sum F_{pq} S^x_{pq}$. For a state average the same terms are taken at root I's Lagrangian densities: root I's RDMs, plus the transition RDMs of z_c, plus the orbital response of z_o (section 7).

The Hessian is d/dy of (8), or equivalently the second derivative of the energy along the parameter-fixed path plus the response. The second view is the one that organizes the work, and it starts with the path.

5 The connection, to second order

At fixed parameters the MOs must stay orthonormal as the basis moves. With $s(x) = C^{\mathsf T} S(x) C - 1$, the symmetric (Löwdin) connection is

$$C(x) = C\, T(x), \qquad T = (1+s)^{-1/2} = 1 - \tfrac{1}{2} s + \tfrac{3}{8} s^2 - \cdots \tag{9}$$

Expanding $s = S^x\, \delta x + \frac{1}{2} S^{xy}\, \delta x\, \delta y$ (MO basis throughout) gives the first and second derivatives of T,

$$\tau^x = -\tfrac{1}{2} S^x, \qquad \tau^{xy} = -\tfrac{1}{2} S^{xy} + \tfrac{3}{8} (S^x S^y + S^y S^x). \tag{10}$$

The tree's gradient uses τ^x only. **The linearized connection $1 - \frac{1}{2} S^x \delta x$ is not orthonormal at second order**, and a Hessian built on it is wrong by terms in $S^x S^y$. Any connection that is orthonormal to second order gives the same final Hessian at a stationary point; the pieces differ. The symmetric one is used here because it is the one the gradient and `response::fock` already assume.

The fixed-parameter second derivative. Write $E(T)$ for the energy (6) with every MO integral transformed by a general, not necessarily orthogonal, T: $h \to T^{\mathsf T} h T$, and T on each index of $(pq|rs)$. Then $\partial E/\partial T_{pq} = 2F_{pq}$ at $T = 1$, and its second derivative along two generators A and B is the energy at the *doubly one-index transformed* integrals,

$$E^{(2)}[A,B] = \sum D_{pq} (A^{\mathsf T} h B + B^{\mathsf T} h A)_{pq} + \tfrac{1}{2} \sum \Gamma_{pqrs}\, (pq|rs)^{[A,B]}, \tag{11}$$

$(pq|rs)^{[A,B]}$ being the sum over every way of applying A to one index and B to another. This is the machinery behind the orbital Hessian, `response::Hessian`, used here with symmetric generators instead of antisymmetric ones. Collecting terms, the second derivative of the energy at fixed λ is

$$\boxed{\begin{aligned} E_{xy} = \underbrace{\sum D^{\mathrm{AO}} h^{xy} + \tfrac{1}{2}\sum \Gamma^{\mathrm{AO}} (\mu\nu|\lambda\sigma)^{xy} + V_{nn}^{xy}}_{\text{explicit}} + 2\sum_{pq} F_{pq}\,\tau_{pq}^{xy} \\ + 2\sum_{pq} \left(F_{pq}^{[x]}\,\tau_{pq}^{y} + F_{pq}^{[y]}\,\tau_{pq}^{x}\right) + E^{(2)}[\tau^x, \tau^y]. \end{aligned}} \tag{12}$$

The second term reduces to $-\operatorname{tr}(WS^{xy})$ plus the $\frac{3}{8}$ terms, the RHF $-WS^{xy}$ plus its second-order correction. The third is the derivative-integral Fock matrix against the first-order connection. The last is the connection acting twice. For RHF these collapse into the S_{ij}^{x} terms of the RHF Hessian's symmetric form.

6 State-specific CASSCF

6.1 The Hessian

$E(x, \lambda)$ is the energy at geometry x and parameters λ along the connection of section 5, with the orbitals

$$C(x, \kappa) = C\,T(x)\,e^{K(\kappa)}\,: \tag{13}$$

the connection first, the rotation second. The order changes the pieces, not the total. Every per-coordinate formula below, and the z-density shortcut of section 7, assumes this one. At the solution $E_\lambda = 0$ at every geometry, so

$$\frac{dE}{dx} = E_x, \qquad \frac{d^2E}{dx\,dy} = E_{xy} + E_{x\lambda}\,\lambda^y, \qquad E_{\lambda\lambda}\,\lambda^y = -E_{\lambda y}, \tag{14}$$

and therefore

$$\boxed{E^{xy} = E_{xy} - E_{x\lambda}\,E_{\lambda\lambda}^{-1}\,E_{\lambda y} = E_{xy} - \lambda^{x\mathsf{T}} E_{\lambda\lambda}\,\lambda^y.} \tag{15}$$

The response term is negative semi-definite when $E_{\lambda\lambda}$ is positive definite, a minimum in the parameters, as for RHF. A state-specific *excited* root is a saddle point in λ: the response term then has no sign, and a conjugate-gradient solve of $E_{\lambda\lambda}\lambda^y = -E_{\lambda y}$ is not guaranteed to converge. E_{xy} is (12). $E_{\lambda\lambda}$ is the CASSCF parameter Hessian (orbital–orbital, orbital–CI, CI–CI), the one the second-order optimizer and the Z-vector already apply. $E_{x\lambda}$ is the right-hand side below. The design checked (15) numerically against a central difference of PySCF's analytic CASSCF gradient.

6.2 The CP-CASSCF right-hand side

$E_{\lambda y}$ is the parameter gradient differentiated at fixed parameters along the connection:

$$\text{orbital:}\quad g_{pq}^{[y]} + \left(g \text{ at the integrals one-index transformed by } \tau^y\right)_{pq}, \qquad g_{pq}^{[y]} = 2(F_{pq}^{[y]} - F_{qp}^{[y]}), \tag{16}$$

$$\text{CI:}\quad 2P(\tilde{H}^y - \langle c|\tilde{H}^y|c\rangle)c, \qquad \tilde{H}^y = H^{[y]} + H[\tau^y], \tag{17}$$

where $H[\tau^y]$ is the active-space Hamiltonian at the τ^y one-index transformed integrals (core included, through h_{eff}). $F^{[y]}$ needs the derivative integrals with all four indices in the occupied and general spaces, contracted with D and Γ: a *derivative J/K build* over the core density and the active density, plus the active-pair derivative integrals $(pt|uv)^y$. These are the CASSCF version of the RHF skeleton Fock matrices (section 2), and like those they are matrices per coordinate, not the scalars the tree's derivative kernels return.

6.3 Solving it: the Z-vector's operator, $3N$ times

$E_{\lambda\lambda}\lambda^y = -E_{\lambda y}$ is the Z-vector equation with a different right-hand side. `response::zvector` (`src/response/zvector/README.md`) already solves it with a nonzero CI part, by the Schur complement on the CI-relaxed orbital Hessian $\tilde{H}$ of `response::RelaxedHessian`:

$$\tilde{H}\,\kappa^y = -g_y - H_{oc}w_y, \qquad w_y = -(H_0 - E)^{-1}P\,y_y, \qquad c^y = -(H_0 - E)^{-1}P\,H'(\kappa^y)\,c + w_y, \quad (18)$$

with (g_y, y_y) the orbital and CI parts of $E_{\lambda y}$. The scaling trap that README documents applies unchanged: its CI rows are *unweighted and unscaled*, so a right-hand side written with the $2P(H-E)c$ convention of (17) enters halved. Assemble the response term as $\Lambda^{\mathsf{T}}E_{\lambda\cdot}$, the solutions against the right-hand sides, not as $-\Lambda^{\mathsf{T}}E_{\lambda\lambda}\Lambda$. The operator's scaling then cancels between the solve and its right-hand side and never reaches the Hessian.

What changes is the count. One $\tilde{H}$ product is J/K builds plus an inner conjugate-gradient CI solve (up to 200 sigmas per root), so $3N$ right-hand sides through the Schur complement nest one iterative solve inside another, $3N$ times. Two remedies, to be measured rather than assumed:

1. **Batch the Schur complement.** All $3N$ share $\tilde{H}$, so block CG applies one product to $3N$ vectors: one batched J/K build and $3N$ CI solves per iteration.

2. **Solve the coupled system unreduced.** PCG on the full (κ, c) vector with the block-diagonal preconditioner $(\operatorname{diag} H_{oo},\ (H_0 - E_0)_{\text{diag}})$, as PySCF solves its CP-MCSCF equations. No inner solve; each iteration is one orbital Hessian product, one $H'(\kappa)c$ and one sigma per right-hand side.

What the J/K builds see. Wang et al. count two J and two K builds per PCG step of TeraChem's CP-MCSCF, on two densities built from the trial vector p: the core–virtual density D_{CV} (only the $p_{\text{core,virt}}$ block) and the generalized active density

$$D_{\text{Gen}} = \begin{pmatrix} 0 & p_{\text{core,act}}(2-\gamma) & 2p_{\text{core,virt}} \\ 0 & 0 & \gamma\, p_{\text{act,virt}} \\ 0 & 0 & 0 \end{pmatrix}$$

(their Eqs. 14 and 15). Only the symmetric parts contribute, but the symmetrized densities are not positive semi-definite, so they cannot be written XX^{T} for DF-K-MO. Their fix is an SVD, $\tilde{D} = U\Sigma V^{\mathsf{T}}$, giving left and right pseudo-coefficients $C_L = U\Sigma^{1/2}$ and $C_R = V\Sigma^{1/2}$, and an exchange build $K(C_L, C_R)$ over the fitted pairs. **The symmetrized $\tilde{D}$ has rank up to $2n_{\text{occ}}$, not n_{occ}.** The unsymmetrized D_{Gen} and D_{CV} have nonzero rows only in the occupied block, so their rank is at most n_{occ} (or n_{core}). They also come factored, with no SVD: in the AO basis $D = C_o\,(D_{o\cdot}C^{\mathsf{T}})$, so $C_L = C_o$ and $C_R = C\,D_{o\cdot}^{\mathsf{T}}$. $K(C_L, C_R)$ followed by $K \to \frac{1}{2}(K + K^{\mathsf{T}})$ is then $K(\tilde{D})$ at half the rank. The SVD is a compression to try, not a requirement. Every density the Hessian adds is of this kind: the $3N$ response densities per CP-CASSCF iteration, the CI responses' transition RDMs, and in the state average the Z-weighted ones. A batched, left/right DF-K over SVD factors is the J/K the Hessian wants.

Since written. `response::zvectors` takes the $3N$ right-hand sides as a block, and solves them column by column, by the batched Schur complement, or by the coupled system unreduced (`src/response/zvector/README.md`, "Many right-hand sides").

7 State-averaged CASSCF

7.1 The Lagrangian, and why its second derivative needs only Z

Root I's energy is not stationary in the SA parameters. The gradient is that of

$$\mathcal{L}(x, \lambda, z) = E_I(x, \lambda) + z \cdot g_{\mathrm{SA}}(x, \lambda), \qquad g_{\mathrm{SA}} = \partial E_{\mathrm{SA}}/\partial\lambda, \tag{19}$$

with $Az = -\partial E_I/\partial\lambda$, $A = \partial g_{\mathrm{SA}}/\partial\lambda$ the SA Hessian: the tree's Z-vector. $\mathcal{L}$ is stationary in λ and z, and the same elimination as the NEVPT2 Hessian's $2n + 1$ form (`docs/nevpt2/nevpt2.tex`, Eq. (hess)) gives

$$\boxed{E_I^{xy} = \mathcal{L}_{xy} + \mathcal{L}_{x\lambda}\lambda^y + \lambda^{x\mathsf{T}}\mathcal{L}_{\lambda y} + \lambda^{x\mathsf{T}}\mathcal{L}_{\lambda\lambda}\lambda^y, \qquad A\,\lambda^x = -\partial_x g_{\mathrm{SA}}.} \tag{20}$$

- λ^x is the response of the *SA* parameters: $3N$ solves with the SA Hessian A, the same operator as the Z-vector, against the SA right-hand sides (16)–(17) at the averaged densities.
- z enters at zeroth order, the gradient's own. No z^x, which would be $3N$ more solves.
- $\mathcal{L}_{xy} = E_{I,xy} + z \cdot \partial_{xy} g_{\mathrm{SA}}$: the fixed-parameter second derivative (12) at root I's densities plus the same at z's, the transition RDMs of z_c and the z_o-rotated densities. These are exactly the densities the gradient already forms.
- $\mathcal{L}_{x\lambda} = \partial_x\partial_\lambda E_I + z \cdot \partial_x\partial_\lambda g_{\mathrm{SA}}$: root I's CP right-hand side, plus the SA Hessian at derivative integrals (and connection) contracted with z.
- $\mathcal{L}_{\lambda\lambda} = \partial_{\lambda\lambda} E_I + z \cdot \partial_{\lambda\lambda} g_{\mathrm{SA}}$: root I's parameter Hessian, plus a **third derivative of E_{SA} contracted once with z**. Only its action $\lambda^{x\mathsf{T}}(\cdot)\lambda^y$ is needed.

The third-derivative term is the new object, and the warning the NEVPT2 Hessian carries applies to it verbatim: **differencing a Hessian-vector product does not give it** on any pair with an orbital part, because the operator is the Jacobian of g only at a stationary point. An analytic version differentiates `response::fock`'s gradient twice: J/K builds at doubly rotated densities, plus the CI legs as transition densities.

Since written. The third-derivative action is `mcscf::third_derivative` (`src/mcscf/third_derivative`), the covector $a \mapsto T(a, v, z)$ for each vector v of a block.

7.2 The rotations among the averaged roots

The SA energy is invariant to rotations among its roots when their weights are equal. Root I's is not. They are fixed by the condition that H_0 stays diagonal in the roots' span, $\langle c_r|H|c_s\rangle = 0$ for $r \neq s$. With $c_r \to c_r + \sum_s \theta_{sr} c_s$, θ antisymmetric, its first derivative gives

$$\theta_{sr}^x = \frac{\mathcal{H}_{rs}^x}{E_r - E_s}, \qquad \mathcal{H}_{rs}^x = \langle c_r|\tilde{H}^x + H'(\kappa^x)|c_s\rangle, \tag{21}$$

the orbital-relaxed coupling between the two roots, which is the derivative-coupling numerator. The CI response outside the span does not enter it, because $H_0 c_s = E_s c_s$ is orthogonal to that response.

In the gradient. At equal weights the rotations are redundant in E_{SA}, and E_I is stationary in them because the roots are eigenvectors. Their multipliers vanish, and the tree projects them away (`src/response/zvector/README.md`). **At unequal weights they are not redundant.** $\partial^2 E_{\mathrm{SA}}/\partial\theta_{rs}^2 = 2(w_r - w_s)(E_s - E_r)$, and they couple to the orbitals through $(w_r - w_s)$ times the r–s transition orbital gradient, so they belong in A and can carry multipliers. A PySCF model of the state average finds:

- **two roots, any weights**: the multiplier of the one rotation is zero to 3×10^{-7} at weights $(0.7, 0.3)$, $(0.9, 0.1)$ and $(0.35, 0.65)$, against a coupling up to 0.16. The projected gradient is exact. This is observed, not proven;
- **three roots, weights** $(0.5, 0.3, 0.2)$: the rotations θ_{01} and θ_{02} again carry no multiplier, but θ_{12} carries 0.07–0.15. The projected gradient is **wrong by up to** 3.3×10^{-3}, 7% of root 2's gradient. With the rotations in A it is exact to 2×10^{-6}.

So the gradient as it then was is right at equal weights and for two roots, and wrong for three or more roots at unequal weights. That case is outside PySCF's analytic SA gradient too, which refuses unequal weights. Until the tree carries the rotations for pairs of unequal weight, it should refuse that case. Partly equal weights, where some rotations are redundant and some are not, were not checked.

In the Hessian. At equal weights the rotations are not in λ, and they add a term of their own to (20). E_I is stationary in θ_{sI}, with $\partial^2 E_I/\partial\theta_{sI}^2 = 2(E_s - E_I)$ and total derivative $2\mathcal{H}^x_{Is}$ along the response. Eliminating θ then gives

$$\boxed{E_I^{xy} \mathrel{+}= \sum_{s\neq I} \frac{\mathcal{H}^x_{Is}\mathcal{H}^y_{sI} + \mathcal{H}^y_{Is}\mathcal{H}^x_{sI}}{E_I - E_s} = 2\sum_{s\neq I} \frac{\mathcal{H}^x_{Is}\mathcal{H}^y_{Is}}{E_I - E_s}.} \tag{22}$$

The factor 2 is not optional. With only one copy of the product, the Hessian is wrong by half this term, 0.02 to 0.24 in the design's checks. With three equal roots the term is as large as the Hessian itself. At unequal weights the rotations are part of λ, their response is inside λ^x, and (20) is complete without (22).

Either way the term **diverges as** $E_s \to E_I$. That is the physics: at a conical intersection the adiabatic energy is not differentiable and has no Hessian. `RelaxedHessian` already drops the other-root part of the CI response for a gap under 10^{-8} hartree. A Hessian must instead refuse, with the gap in the message, below a threshold on $|\mathcal{H}^x_{Is}|/|E_I - E_s|$ that is far larger. The tree's weights are recovered from a solution and need not be equal, so both paths are needed.

Since written. Every pair of unequal weight is a parameter of A with a multiplier of its own (`src/response/zvector/README.md`, "The rotations among the roots"), and the equal-weight term (22), with the rotations (21), is `mcscf::root_rotation` (`src/mcscf/root_rotation`).

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